Single Image Haze Removal Using Dark Channel Prior

Thomas Markussen and Preben Birkeland Wassbakk

IKT452-G 26V

Computer Vision

University of Agder

Grimstad, Spring 2026

Abstract

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List of Abbreviations

Abbreviation	Definition
AOD-Net	All-In-One Dehazing Network
ASM	Atmospheric Scattering Model
CAP	Color Attenuation Prior
CNN	Convolutional Neural Network
DCP	Dark Channel Prior
DCPDN	Densely Connected Pyramid Dehazing Network
DFF	Dense Feature Fusion
GAN	Generative Adversarial Network
ICCV	International Conference on Computer Vision
MSBDN	Multi-Scale Boosted Dehazing Network
O-HAZE	Outdoor Hazy Image Dataset
PSNR	Peak Signal-to-Noise Ratio
RESIDE	REalistic Single Image DEhazing dataset
RGB	Red, Green, Blue
SSIM	Structural Similarity Index Measure
SOTS	Synthetic Objective Testing Set

1 Introduction

1.1 Acknowledgments

AI tools (ChatGPT) were used only for language improvements and small suggestions to the structure. All model training, evaluation, analysis and conclusions were done by us.

1.2 Workload share

2 Background/Related work

2.1 Single Image Haze Removal Using Dark Channel Prior (Main paper)

Single-image haze removal is an important topic in computer vision because haze reduces visibility, lowers contrast, and distorts scene colors. These degradations make it harder not only for humans to interpret images, but also for computer vision systems to perform tasks such as object detection, segmentation, and scene understanding. The paper Single Image Haze Removal Using Dark Channel Prior by He, Sun, and Tang became one of the most influential early works in this area because it provided a practical solution for recovering a clear image from a single hazy input.

The method is built on the atmospheric scattering model, where the observed hazy image is assumed to be a combination of attenuated scene radiance and airlight caused by particles in the atmosphere. Since both the transmission map and the atmospheric light are unknown, the problem is ill-posed when only one image is available. The main contribution of the paper is the introduction of the dark channel prior, a statistical observation that in most local patches of outdoor haze-free images, at least one color channel contains pixels with very low intensity. By exploiting this property, the method can estimate haze thickness and then recover the underlying clear image.

A major reason this paper became so important is that it showed how a carefully designed image prior could solve a difficult restoration problem without requiring multiple images, special sensors, or additional scene information. The approach was simple, intuitive, and effective, which made it highly influential in later dehazing research.

The paper also notes that the estimated transmission can be used to obtain a depth map as a byproduct. This further increased its relevance within computer vision, since the method was not only useful for image restoration, but also provided information about

scene structure.

Figure 1: Haze removal using a single image. (a) Hazy image. (b) After haze removal. (c) Depth map

From a research perspective, this paper represents a landmark prior-based dehazing method. It established a strong baseline for later work and became a common reference point for evaluating new methods. At the same time, it also had limitations. Because the dark channel assumption does not always hold in bright regions, such as skies or light-colored objects, the method can fail in these areas or introduce artifacts. These weaknesses motivated subsequent papers to refine transmission estimation, improve robustness, and eventually move toward learning-based approaches [1].

2.2 AOD-Net: All-in-One Dehazing Network (2017)

One important direction of later dehazing research was the move away from methods that estimate the transmission map and atmospheric light in separate steps. AOD-Net represents this shift clearly. The paper argues that the traditional pipeline can accumulate errors, since inaccuracies in transmission estimation and atmospheric light estimation may be amplified when combined in the final image restoration stage. Instead of following that process, AOD-Net reformulates the atmospheric scattering model and learns a direct

mapping from a hazy image to a clean image using a lightweight convolutional neural network.

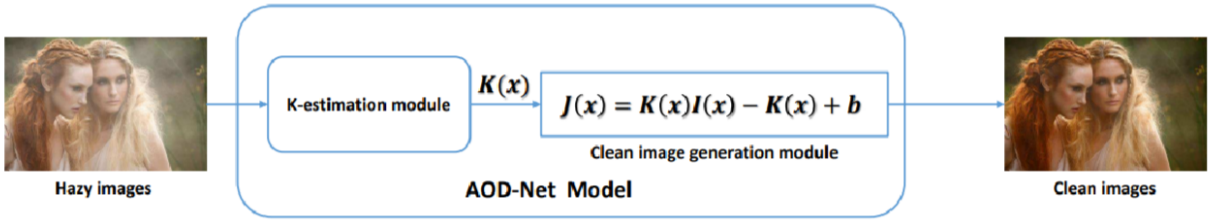

Figure 2: The diagram of AOD-Net

This was an important development because it changed how the dehazing problem was framed. Rather than treating image restoration mainly as a consequence of estimating physical parameters first, the paper treats restoration itself as the main learning target.

The paper reports improved PSNR and SSIM compared with several earlier methods, including the original dark channel prior approach, on synthetic benchmark datasets. It also shows that AOD-Net can be combined with Faster R-CNN and improve object detection performance on hazy images. This gave the paper added relevance, since it suggested that dehazing should not only be evaluated as a low-level image enhancement task, but also in terms of how it affects later vision tasks [2].

2.3 Densely Connected Pyramid Dehazing Network (DCPDN) (2018)

A later and more structured development is presented in DCPDN. While AOD-Net simplified the dehazing problem through an all-in-one formulation, DCPDN instead kept the atmospheric scattering model explicit and embedded it directly into a deep learning framework. In this way, the method jointly optimizes transmission estimation, atmospheric light estimation, and final image restoration. The paper argues that previous learning-based methods were still limited by inaccurate transmission maps and weak interaction between different parts of the dehazing process. To address these weaknesses, DCPDN introduces a densely connected architecture with pyramid pooling, edge-preserving loss, and joint optimization of transmission, atmospheric light, and image restoration. This gives the method a stronger balance between physical modeling and learned feature extraction.

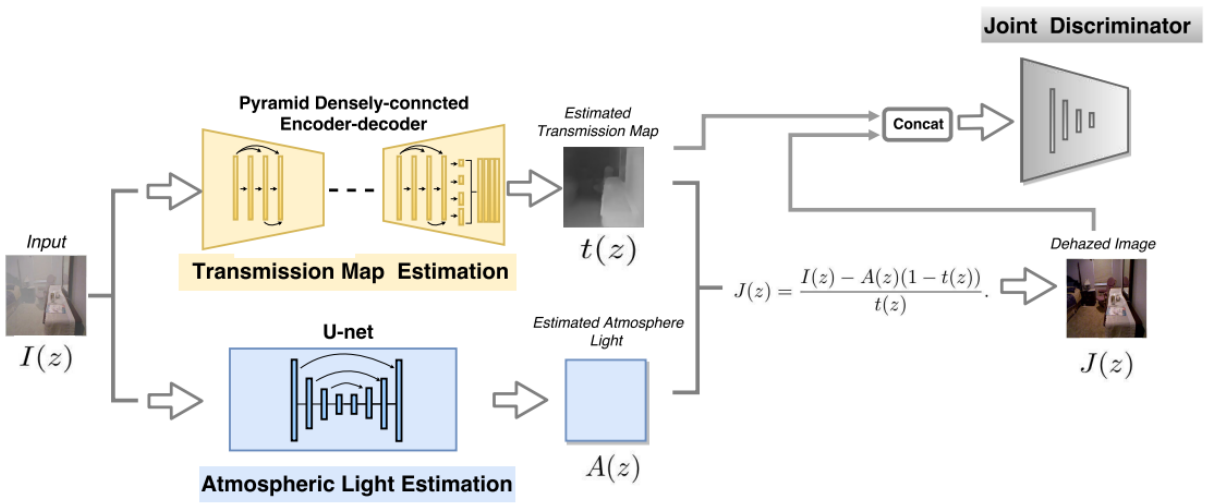

Figure 3: Overview of the DCPDN architecture. The model estimates the transmission map and atmospheric light separately, then combines them through the atmospheric scattering model to produce the final dehazed image.

The paper reports that these components improve both the estimated transmission maps and the restored images. Compared with earlier methods, including the original dark channel prior paper and AOD-Net, DCPDN achieves stronger results on synthetic and real datasets. The authors also emphasize that the method preserves sharper contours, avoids some of the darker or hazier outputs seen in previous methods, and produces results that are visually closer to the ground truth. Because of this, DCPDN can be seen as a bridge between two research directions. It does not abandon the physical haze model, but it also does not rely only on hand-crafted priors. Instead, it uses deep learning to improve how the physical model is estimated and optimized [3].

2.4 Color-Constrained Dehazing Model (2020)

The paper focuses on an important weakness in earlier dehazing methods: unrealistic color reconstruction. The paper argues that the commonly used atmospheric scattering model assumes global uniform atmospheric light and does not sufficiently control the colors in the final dehazed image. As a result, previous methods may remove haze while still introducing unnatural colors or visible color distortion.

GridDehazeNet [18] Our result

Figure 4: Visual comparison of the Color-Constrained Dehazing Model with GridDehazeNet on a challenging hazy image. The proposed method produces a clearer result with less color distortion and more natural color.

To address this, the authors propose a modified dehazing model that uses local atmospheric light instead of a single global value and adds explicit color constraints based on the distribution of colors found in haze-free images. This allows the method to improve not only visibility, but also color realism. The reported experiments show that it outperforms several previous methods, including AOD-Net and DCPDN, on HazeRD, RESIDE, and O-HAZE in terms of PSNR and SSIM. This makes it a useful final paper in the progression, since it shows that later dehazing research was not only about removing more haze, but also about recovering more realistic scene color and appearance [4].

3 Approach/Methodology

3.1 Problem formulation

Outdoor images captured in hazy conditions often suffer from reduced visibility, lower contrast, and distorted colors, which can negatively affect both human interpretation and computer vision systems. Although several single-image dehazing methods have been pro-

posed, it is unclear how well influential dehazing approaches from different stages of the literature perform when applied to a new practical setting. This project investigates how the original Dark Channel Prior method and three later successor papers perform on a selected real-world dehazing problem, with emphasis on image quality and practical usefulness.

Hypothesis: *Later learning-based dehazing methods will outperform the original Dark Channel Prior method in the selected application setting, both in terms of quantitative image quality and visual realism.*

3.2 Datasets

This project uses the RESIDE benchmark as the main synthetic dataset [5] and O-HAZE as an additional evaluation dataset [6]. For training, the learning-based methods use training images from RESIDE, including the indoor training set (ITS) and the outdoor training set (OTS)[5]. These training sets provide synthetic hazy images together with corresponding haze-free ground truth images, making them suitable for supervised learning of single-image dehazing methods.

For testing, RESIDE SOTS is used as the main benchmark. RESIDE SOTS is one of the most widely used benchmarks for single-image dehazing and provides synthetic hazy images together with corresponding haze-free ground truth images. This makes it suitable for quantitative evaluation using metrics such as PSNR and SSIM. Since it contains both indoor and outdoor images, it allows evaluation on a broader set of hazy scenes.

The O-HAZE dataset is used to evaluate how well the compared methods perform under more realistic haze conditions. Unlike RESIDE, where haze is generated synthetically from clean images, O-HAZE contains real outdoor hazy and haze-free image pairs captured under controlled conditions. This makes it useful for examining whether methods that perform well on synthetic benchmarks also produce visually realistic results on real hazy images. The combination of RESIDE and O-HAZE provides a balanced evaluation setup. RESIDE supports systematic quantitative comparison on a larger benchmark, while O-HAZE makes it possible to assess generalization and visual quality on real outdoor scenes.

3.3 Compared Methods

Four methods are compared in this project: the main paper, *Single Image Haze Removal Using Dark Channel Prior*, and three later successor methods, *AOD-Net*, *Densely Connected Pyramid Dehazing Network (DCPDN)*, and *Color-Constrained Dehazing Model*. These methods were selected because they represent different stages in the development of single-image dehazing, from prior-based restoration to later learning-based and model-aware approaches. They together provide a progression from the original dark channel prior method to methods that aim to improve restoration quality, joint optimization, and color realism.

3.4 Evaluation Metrics

The compared methods are evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). PSNR is used to measure reconstruction quality at the pixel level, while SSIM is used to evaluate the structural similarity between the dehazed image and the corresponding ground-truth image. In addition to these quantitative metrics, qualitative comparison is also included in order to assess visual realism, color reconstruction, contrast, and possible artifacts in the dehazed outputs.

4 Implementation

5 Analysis

6 Conclusion

References

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